

Normal Distribution, Pt 2

CSU, Chico Math 350

2026-04-27

outline

Normal Random Variable

Inverse Normal

References

outline

Normal Random Variable

Inverse Normal

References

Recap

Last time, we saw the properties of the Standard Normal Random Variable, with mean 0 and variance 1. While useful, its properties only apply for that specific μ and σ^2 .

Recall that the *normal random variable* X has mean μ , standard deviation σ , and PDF:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Using the tools we've seen in past chapters, we can solve problems for values of μ and σ beyond just the Standard Normal case.

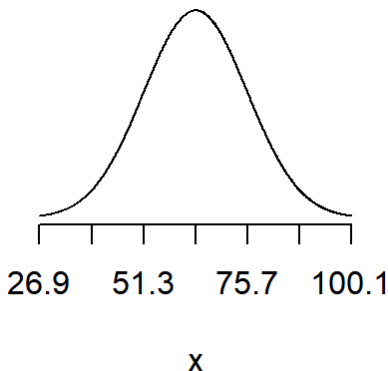
Example 1

A normally distributed population of lemming body weights has a mean of 63.5 g and a standard deviation 12.2 g.

First, define the random variable:

Let X be the weight of a lemming in grams.

Then, draw a picture of the distribution:



Example 1

From here, we can use both theoretical calculations (using `pnorm()`) and simulation (using `rnorm()`) to answer questions related to the weight of lemmings in this population.

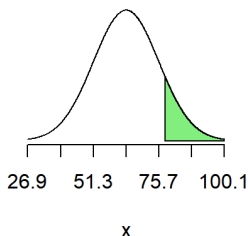
For the problems that follow, we can define our simulated sample using:

```
body.weights <- rnorm(10000, 63.5, 12.2)
```

This will generate a sample of size 10000 from the Normal distribution with mean $\mu = 63.5$ and standard deviation $\sigma = 12.2$ (recall that `rnorm` takes standard deviation as an argument, not variance).

Example 1

1) What proportion of this population is 78.0 g or larger?



Theoretical

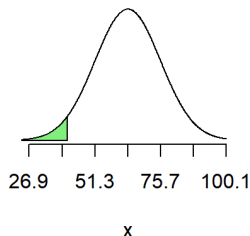
```
1-pnorm(78, 63.5, 12.2)  
## [1] 0.1173134
```

Simulation

```
mean(body.weights > 78)  
## [1] 0.1201
```

Example 1

2) What is the probability of choosing at random from this population a weight smaller than 41g?



Theoretical

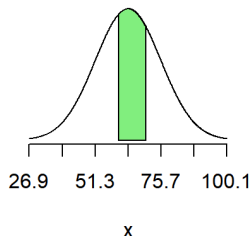
```
pnorm(41, 63.5, 12.2)
## [1] 0.03257246
```

Simulation

```
mean(body.weights < 41)
## [1] 0.0317
```

Example 1

3) What is the probability of choosing at random from this population a weight between 60 and 70 g?



Theoretical

```
pnorm(70, 63.5, 12.2)-pnorm(60, 63.5, 12.2)  
## [1] 0.3158093
```

Simulation

```
mean(body.weights < 70 & body.weights > 60)  
## [1] 0.312
```

Exercise 1: Tarantulas

According to a recent study, the carapace length for adult males of a certain species of tarantula are normally distributed with a mean of 17.45 mm and a standard deviation of 1.85 mm. Answer these questions using both theoretical values (the R function `pnorm()`, and simulation (using `rnorm()`).

- 1) Define the random variable X for this distribution.
- 2) What is the probability that the length of a carapace is between 16 mm and 18 mm?
- 3) Would a tarantula that had a carapace longer than 21 mm be unusual, and why? Include the probability in your answer.

outline

Normal Random Variable

Inverse Normal

References

Concept

It is often of interest to calculate a quantile of the normal distribution. For instance, maybe we want to know the score you would need on the SAT exam to be in the top 10th percentile. For this sort of problem we would want to use the R function `qnorm(p,mu,sigma)`, where p is the area to the **left** of a certain value of interest.

A *quantile* divides the range of a probability distribution into intervals of equal probability.

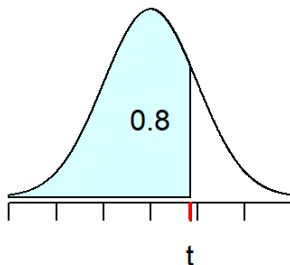
Example 2

Let's look at those lemmings again. Recall the weight of a lemming can be described as $X \sim N(63.5, 12.1^2)$. What lemming weight corresponds to the 80th percentile?

First, translate the question into mathematical notation:

Find t such that $P(X < t) = 0.8$

Then, draw a picture of the distribution:



x

Example 1

As before, we can use both theoretical calculations (using `qnorm()` this time) and simulation (using `rnorm()`) to answer questions related to the quantiles.

Theoretical:

```
qnorm(.8, 63.5, 12.2)
## [1] 73.76778
```

Simulation:

```
lemming <- rnorm(10000, 63.5, 12.2)
quantile(lemming, .8)
##      80%
## 73.79352
```

Exercise 2: Tarantulas, Too

Reconsider the tarantula example above. What carapace length corresponds to the top 10th percentile? Note the different wording here compared to the previous case.

- 1) Define the question in mathematical notation.
- 2) Answer using theoretical values.
- 3) Answer using simulation.

outline

Normal Random Variable

Inverse Normal

References

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.